

Maclaurin Series & Convergence of Series

ANSWER KEY



Constructing a Maclaurin series

- Find the first four non-zero terms of the Maclaurin series for $f(x) = e^x$, using
$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \dots$$
Since every derivative of e^x equals e^x , and $e^0 = 1$: all derivatives at 0 equal 1. So
$$e^x \approx 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} = 1 + x + \frac{x^2}{2} + \frac{x^3}{6}.$$
- Find the Maclaurin series for $f(x) = \sin x$ up to the term in x^5 .
 $f(0) = 0, f'(0) = \cos 0 = 1, f''(0) = -\sin 0 = 0, f'''(0) = -\cos 0 = -1, f^{(4)}(0) = \sin 0 = 0,$
 $f^{(5)}(0) = \cos 0 = 1$. Series: $\sin x \approx x - \frac{x^3}{3!} + \frac{x^5}{5!} = x - \frac{x^3}{6} + \frac{x^5}{120}.$

Using known series to build new ones

- Using the Maclaurin series for e^x , find the Maclaurin series for e^{-2x} up to the term in x^3 .
Substitute $-2x$ for x in $e^x \approx 1 + x + \frac{x^2}{2} + \frac{x^3}{6}$:
$$e^{-2x} \approx 1 + (-2x) + \frac{(-2x)^2}{2} + \frac{(-2x)^3}{6} = 1 - 2x + 2x^2 - \frac{4x^3}{3}.$$
- Given the Maclaurin series $\sin x \approx x - \frac{x^3}{6}$ and $\cos x \approx 1 - \frac{x^2}{2}$, find the Maclaurin series for $\tan x$ up to the term in x^3 , using $\tan x = \frac{\sin x}{\cos x}$ and polynomial division.
$$\frac{x - x^3/6}{1 - x^2/2} \approx \left(x - \frac{x^3}{6}\right)\left(1 + \frac{x^2}{2}\right) \text{ (using } \frac{1}{1-u} \approx 1 + u \text{ for small } u) = x + \frac{x^3}{2} - \frac{x^3}{6} + O(x^5) = x + \frac{x^3}{3}.$$

Using a Maclaurin series to approximate a value

- Use the Maclaurin series for e^x up to the x^3 term to approximate $e^{0.1}$, to four decimal places, and compare with the true value $e^{0.1} \approx 1.10517$.
$$e^{0.1} \approx 1 + 0.1 + \frac{(0.1)^2}{2} + \frac{(0.1)^3}{6} = 1 + 0.1 + 0.005 + 0.0001\bar{6} \approx 1.1052.$$
 This is very close to the true value 1.10517, with an error of about 0.00003.

Tests for convergence of a series

- Use the ratio test to determine whether $\sum_{n=0}^{\infty} \frac{n}{2^n}$ converges.
Ratio test: $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \frac{(n+1)/2^{n+1}}{n/2^n} = \lim_{n \rightarrow \infty} \frac{n+1}{2n} = \frac{1}{2}$. Since $\frac{1}{2} < 1$, the series converges.
- Use the ratio test to determine whether the Maclaurin series for e^x , $\sum_{n=0}^{\infty} \frac{x^n}{n!}$, converges for all real x .
Ratio test: $\lim_{n \rightarrow \infty} \left| \frac{x^{n+1}/(n+1)!}{x^n/n!} \right| = \lim_{n \rightarrow \infty} \frac{|x|}{n+1} = 0$ for any fixed x . Since the limit is $0 < 1$ regardless of x , the series converges for all real values of x .

Determining the radius of convergence

- 8 Use the ratio test to find the radius of convergence of the power series $\sum_{n=1}^{\infty} \frac{x^n}{n}$.
Ratio test: $\lim_{n \rightarrow \infty} \left| \frac{x^{n+1}/(n+1)}{x^n/n} \right| = \lim_{n \rightarrow \infty} \frac{n|x|}{n+1} = |x|$. The series converges when $|x| < 1$, so the radius of convergence is $R = 1$.
- 9 Find the radius of convergence of $\sum_{n=0}^{\infty} n! x^n$, and explain what this tells us about where the series is useful.
Ratio test: $\lim_{n \rightarrow \infty} \left| \frac{(n+1)! x^{n+1}}{n! x^n} \right| = \lim_{n \rightarrow \infty} (n+1)|x| = \infty$ for any $x \neq 0$. Since this limit is never less than 1 (except at $x = 0$), the radius of convergence is $R = 0$ — the series only converges at the single point $x = 0$, making it useless as an approximation anywhere else.
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Comparing a Maclaurin approximation to the exact function

- 10 Using the Maclaurin series $\cos x \approx 1 - \frac{x^2}{2} + \frac{x^4}{24}$, approximate $\cos(0.3)$ to four decimal places, and compare with the true value $\cos(0.3) \approx 0.9553$.
 $\cos(0.3) \approx 1 - \frac{(0.3)^2}{2} + \frac{(0.3)^4}{24} = 1 - 0.045 + 0.0003375 \approx 0.9553$. This matches the true value to four decimal places, showing the Maclaurin approximation is highly accurate for small x .
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Maclaurin series for $\ln(1+x)$

- 11 Find the Maclaurin series for $f(x) = \ln(1+x)$ up to the term in x^4 , using
 $f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4$.
 $f(0) = \ln 1 = 0$. $f'(x) = (1+x)^{-1}$, so $f'(0) = 1$. $f''(x) = -(1+x)^{-2}$, so $f''(0) = -1$. $f'''(x) = 2(1+x)^{-3}$, so $f'''(0) = 2$. $f^{(4)}(x) = -6(1+x)^{-4}$, so $f^{(4)}(0) = -6$. Series:
 $\ln(1+x) \approx x - \frac{x^2}{2} + \frac{2x^3}{6} - \frac{6x^4}{24} = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4}$.
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Determining the interval of convergence using the ratio test

- 12 Find the radius of convergence of the Maclaurin series for $\ln(1+x)$, $\sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^n}{n}$, using the ratio test.
Ratio test: $\lim_{n \rightarrow \infty} \left| \frac{x^{n+1}/(n+1)}{x^n/n} \right| = \lim_{n \rightarrow \infty} \frac{n|x|}{n+1} = |x|$. The series converges when $|x| < 1$, so the radius of convergence is $R = 1$ (consistent with the earlier result found for a structurally similar series).
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Combining Maclaurin series through multiplication

- 13 Using the Maclaurin series $e^x \approx 1 + x + \frac{x^2}{2}$ and $\frac{1}{1-x} \approx 1 + x + x^2$ (valid for small x), find the Maclaurin series for $\frac{e^x}{1-x}$ up to the term in x^2 , by multiplying the two series together.
Multiply term by term, keeping only terms up to x^2 :
 $(1 + x + \frac{x^2}{2})(1 + x + x^2) = 1 + x + x^2 + x + x^2 + \frac{x^2}{2} + O(x^3) = 1 + 2x + \frac{5x^2}{2} + O(x^3)$. So
 $\frac{e^x}{1-x} \approx 1 + 2x + \frac{5}{2}x^2$.

Using differentiation to find a Maclaurin series

- 14** Given the Maclaurin series $\frac{1}{1-x} \approx 1 + x + x^2 + x^3 + \dots$ for $|x| < 1$, differentiate both sides term by term to find the Maclaurin series for $\frac{1}{(1-x)^2}$ up to the term in x^2 .
Differentiating $\frac{1}{1-x}$ gives $\frac{1}{(1-x)^2}$ on the left. Differentiating the right side term by term:
 $\frac{d}{dx}(1 + x + x^2 + x^3 + \dots) = 1 + 2x + 3x^2 + \dots$. So $\frac{1}{(1-x)^2} \approx 1 + 2x + 3x^2$.

Estimating an error bound in a Maclaurin approximation

- 15** The Maclaurin series for $\sin x$ truncated after the x^3 term is $\sin x \approx x - \frac{x^3}{6}$. Use this to approximate $\sin(0.2)$ to four decimal places, and find the absolute error compared to the true value $\sin(0.2) \approx 0.19867$.
 $\sin(0.2) \approx 0.2 - \frac{(0.2)^3}{6} = 0.2 - \frac{0.008}{6} = 0.2 - 0.001333 \approx 0.19867$. Absolute error:
 $|0.19867 - 0.19867| \approx 0.00000$ (to four decimal places), showing that even a low-order Maclaurin approximation is extremely accurate for small values of x close to 0.
- 16** Find the first three non-zero terms of the Maclaurin series for $f(x) = (1+x)^{1/2}$, using the binomial series $(1+x)^n \approx 1 + nx + \frac{n(n-1)}{2}x^2$ with $n = \frac{1}{2}$.
Substituting $n = \frac{1}{2}$: $(1+x)^{1/2} \approx 1 + \frac{1}{2}x + \frac{\frac{1}{2}(-\frac{1}{2})}{2}x^2 = 1 + \frac{1}{2}x - \frac{1}{8}x^2$.
- 17** Use the Maclaurin series for e^x up to the x^4 term to approximate $\lim_{x \rightarrow 0} \frac{e^x - 1 - x}{x^2}$, and check that your answer matches the value obtained using L'Hopital's rule.
Using $e^x \approx 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \dots$: $e^x - 1 - x \approx \frac{x^2}{2} + \frac{x^3}{6} + \dots$. Dividing by x^2 : $\frac{e^x - 1 - x}{x^2} \approx \frac{1}{2} + \frac{x}{6} + \dots$, which tends to $\frac{1}{2}$ as $x \rightarrow 0$. Using L'Hopital's rule twice on the original limit (since it is initially $\frac{0}{0}$) gives $\lim_{x \rightarrow 0} \frac{e^x - 1}{2x} = \frac{e^0}{2} = \frac{1}{2}$, confirming the same result.
- 18** This question has three parts. (a) Find the Maclaurin series for $f(x) = \cos(x^2)$ up to the term in x^8 , by substituting x^2 into the known series $\cos u \approx 1 - \frac{u^2}{2} + \frac{u^4}{24}$. (b) Use your result to write down an approximation for $\int_0^1 \cos(x^2) dx$ as a sum of simple fractions (this integral has no elementary closed form, so series approximation is genuinely useful here). (c) Evaluate this approximation as a single decimal, to three decimal places.
(a) Substituting $u = x^2$: $\cos(x^2) \approx 1 - \frac{(x^2)^2}{2} + \frac{(x^2)^4}{24} = 1 - \frac{x^4}{2} + \frac{x^8}{24}$. (b) Integrating term by term:
 $\int_0^1 \left(1 - \frac{x^4}{2} + \frac{x^8}{24}\right) dx = \left[x - \frac{x^5}{10} + \frac{x^9}{216}\right]_0^1 = 1 - \frac{1}{10} + \frac{1}{216}$. (c) $1 - 0.1 + 0.00463 \approx 0.905$.